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Dual-space high-frequency learning for transformer-based MRI super-resolution

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A R T I C L E I N F O

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Single image super-resolution (SISR)
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A B S T R A C T

Background and Objective: Magnetic resonance imaging (MRI) can provide rich and detailed high-contrast information of soft tissues, while the scanning of MRI is time-consuming. To accelerate MR imaging, a variety of Transformer-based single image super-resolution methods are proposed in recent years, achieving promising results thanks to their superior capability of capturing long-range dependencies. Nevertheless, most existing works prioritize the design of transformer attention blocks to capture global information. The local high-frequency details, which are pivotal to faithful MRI restoration, are unfortunately neglected.
Methods: In this work, we propose a high-frequency enhanced learning scheme to effectively improve the awareness of high frequency information in current Transformer-based MRI single image super-resolution methods. Specifically, we present two entirely plug-and-play modules designed to equip Transformer-based networks with the ability to recover high-frequency details from dual spaces: 1) in the feature space, we design a high-frequency block (**Hi-Fe block**) paralleled with Transformer-based attention layers to extract rich high-frequency features; while 2) in the image intensity space, we tailor a high-frequency amplification module (**HFA**) to further refine the high-frequency details. By fully exploiting the merits of the two modules, our framework can recover abundant and diverse high-frequency information, rendering faithful MRI super-resolved results with fine details.
Results: We integrated our modules with six Transformer-based models and conducted experiments across three datasets. The results indicate that our plug-and-play modules can enhance the super-resolution performance of all foundational models to varying degrees, surpassing the capabilities of existing state-of-the-art single image super-resolution networks.
Conclusion: Comprehensive comparison of super-resolution images and high-frequency maps from various methods, clearly demonstrating that our module possesses the capability to restore high-frequency information, showing huge potential in clinical practice for accelerated MRI reconstruction.

1. Introduction

Magnetic resonance imaging (MRI) is a fundamental medical imaging technique for generating images with better soft-tissue contrast using diverse pulse sequences. However, due to inherent limitations in MRI systems [1] and crepitations within particular regions of the body, e.g., the chest during respiration [2], acquiring high-resolution (HR) MR images poses great challenges in clinics [3]. Furthermore, patients who undergo MR scan are required to remain still for long periods despite the loud scanner noise, as even slight movements can result in motion artifacts that severely degrade image quality [4]. Therefore, accelerating MRI acquisition is necessary for clinical practice. Single image super-resolution (SISR) techniques efficiently accelerate MR image acquisition by restoring abundant details and textures from a single low-resolution (LR) image obtained during a short scanning process, and thus facilitate various crucial downstream tasks, including MRI segmentation [5–7], registration [8–10], tumor detection [11–13], etc.

Initially, some classic algorithms, such as examples-aided redundant dictionary learning [14], coupled dictionaries [15] and multi-atlas patchmatch [16], have been employed for MR image super-resolution. However, these methods exhibited poor performance primarily due to their constrained model characteristics. Nowadays, various deep learning networks [17–20] have been applied in SISR as the thriving technology. Because the convolutional layers are effective in capturing lo-

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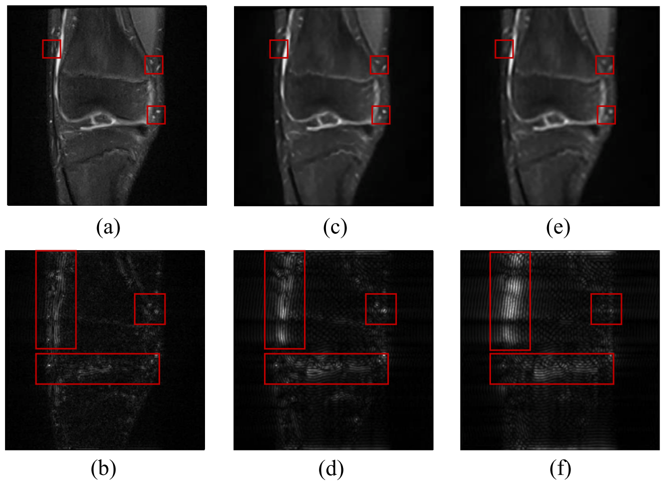

Fig. 1. Illustration on the poor recovery of HF information by existing SISR method and the contribution of our modules. (a,b) are the high-resolution Knee MR image and its HF map; (c,d) are SISR result and the corresponding HF map from the incorporation of our modules (Hi-Fe block and HFA) and Restormer; (e,f) are SISR result and its HF map from Restormer. The red rectangles in images highlight the large differences between the SISR result and the HF map of two methods. It can be seen that after optimization with our module, (d) exists fewer artifacts and checkerboard effects than (f), and is closer to the HF map (b) of the ground truth.

cal textures efficiently, enabling accurate predictions of high-resolution image details. Previous studies have introduced residual connections and skip connections to facilitate the stacking of convolutional neural networks (CNN) [21–23]. Further, Transformer-based models [24–26] show outstanding performance in generating macroscopical structure due to their intrinsic capability in encoding long-range dependencies and in learning global feature representations [4]. For example, Restormer [24] introduced multi-Dconv head transposed attention module and gated-Dconv feed-forward network to capture long-range pixel interactions. However, Transformer-based models underperform in recovering high frequency (HF) information, as shown in Fig. 1, there is a significant discrepancy between the HF map from the output of Restormer (b) and the original map (f). The reason is that Transformer-based networks cannot effectively exploit local structure by modeling fragmented image patch dependencies [27]. Although combining CNN with transformer can be beneficial, it is acknowledged that this combination can increase complexity and will consume significant memory resources.

To solve the challenges outlined above, we present a groundbreaking approach for enhancing high-frequency information learning, encompassing two fully plug-and-play modules that effectively empower Transformers to recover high-frequency details from a dual perspective. Firstly, in the deep feature space, we develop Hi-Fe blocks parallel with backbone attention blocks, allowing the image embeddings to additionally restore HF representation. Extracting HF features using this module significantly reduces the training complexity and computational costs of the network. Then, in the image intensity space, we introduce the High-Frequency Amplification (HFA) module. This module, which can be integrated on top of any Transformer-based backbone network, enhances the recovery of critical high-frequency (HF) details. The HFA module extracts HF information via the Fourier Transform and then amplifies these details in coarse images to improve the overall output. In summary, Hi-Fe block and HFA offer a solution for restoring HF information for Transformer-based networks, thereby harnessing the potential of the transformer in the MR SISR tasks. The contributions of our paper are summarized as follows:

- We delve into the limitations of the high-frequency modeling in existing Transformer-based MRI SISR networks, and we introduce an innovative high-frequency enhancement learning scheme, resulting in significantly improved super-resolution images with exquisite fine details.
- We propose two completely plug-and-play modules, the Hi-Fe block, and HFA, to capture rich and diverse high-frequency clues from dual information spaces. The former decently extracts high-frequency features from the deep feature space, while the latter effectively retrieves and amplifies high-frequency details from the image intensity space.
- Comprehensive experimental results on three datasets demonstrate that the proposed method significantly enhances the performance of existing state-of-the-art SISR methods without additional modification of their architectures.

2. Related work

2.1. Single image super-resolution (SISR)

SISR techniques primarily focus on generating high-resolution (HR) images from single low-resolution (LR) images obtained through short-duration scans. Traditional SISR methods can be divided into two categories: 1) interpolation-based methods (e.g., bilinear, bicubic, and nearest); and 2) reconstruction-based methods (e.g., POCS [28] and IBP [29]). Traditional SISR methods usually introduce blurred edges and blocking artifacts in SR images, particularly when the upsampling factor becomes increasingly large, impairing the accuracy of diagnostic interpretation made by clinicians.

Nowadays, deep learning has been increasingly applied in SISR as a thriving technology, and these models can be divided in to two categories: CNN-based models and Transformer-based models. The first CNN-based SISR model, SRCNN [30], has three convolution layers with different kernel sizes to accomplish feature extraction, nonlinear mapping, and final HR image reconstruction. Goyal et al. [31] integrated multiscale convolutional neural networks with wavelet analysis and weighted least squares optimization, facilitating the learning of features with multiple orientations. Tian et al. [32] employed parallel heterogeneous architectures to enhance channel relations and low-frequency structure information, preserving original information with a multilevel enhancement mechanism. CSN [33] divided hierarchical features into two branches, the residual branch to encourage feature reuse and the dense branch to explore new features, while employing merge-and-run mapping to enhance information integration between branches. ENCLN [34] incorporated efficient non-local attention and sparse aggregation methods to approximate exponential functions with linear computational complexity. Nonetheless, the CNN-based networks strictly integrate features in sequence from local to global, and they also face challenges in retaining sufficient local spatial information within their deep features. These limitations result in distortions in the overall image structure, leading to the generation of erroneous structural information.

2.2. SISR with transformer

Transformer has demonstrated powerful performance across numerous image processing and analysis tasks [35,36,10]. Particularly in SISR, Transformer models also offer a promising solution due to their capacity for encoding long-range dependencies and efficient feature representation learning. Both the shallow and deep features in Transformer-based models encompass a blend of local and global information, effectually improve the accuracy of image super resolution [37]. Güngör et al. [25] proposed TranSMS based on system matrix, including vision transformer module, dense convolutional module, and data-consistency module. Liang et al. [26] introduced SwinIR for image super resolution, consisting of three parts: shallow feature extraction, deep feature extraction, and high-quality image reconstruction. Each residual Swin Transformer block in SwinIR has several Swin Transformer layers together with residual connection. Wang et al. [38] proposed Omni-SR with omni self-attention paradigm for simultaneous spatial and channel interactions to lift the restriction of effective receptive field. While Transformer-based models exhibit proficiency in enhancing the overall quality and restoring the global structure (LF part) of SR images, they often fall short in recovering fine details and edge information in the images. The reason is that the Transformer-based methods cannot effectively exploit local structural information (HF part of images), as they divide the image into a sequence of patches and employ self-attention mechanisms to capture their dependencies [27].

To relieve the transformer's inherent deficiency in restoring HF information, some studies integrate specialized CNN structures into Transformer-based models for HF feature extraction. Lu et al. [39] presented ESRT, which contains a CNN and transformer backbone for deep feature extraction, and it models long-term dependencies between similar local image regions. Li et al. [40] combined transformer and CNN structure in CRAFT, which can capture both LF information and HF details. However, the combination of a CNN and Transformer-based network is excessively intricate and resource-intensive. This complexity can lead to high memory consumption and computational demands, potentially hindering efficient model training and deployment.

2.3. Super resolution with high frequency

The frequency domain information of an image encompasses the distribution and attributes of distinct frequency components within the image. These components consist of low-frequency (LF) and high-frequency (HF) components, characterized by their amplitude and phase features. HF components correspond to rapid spatial frequency changes within an image, which means that HF information reflects sharp fluctuations in brightness values at different locations, typically occurring in localized areas of the image, such as edges, textures, and details [41]. Extracting the HF information from the image and incorporating it into the Transformer-based networks can enhance their ability in learning the local features. Decomposing the complex image into more specific and manageable HF and LF information. This strategy can be summarized as "divide-and-conquer" which is widely implemented in SISR [42–44].

Zhang et al. [45] first used a Gaussian filter to decompose the low dose computed tomography image to HF and LF components. Subsequently, they extracted features from these components, effectively promoting the restoration of high-quality LDCT images with the assistance of piece-wise reconstruction. Gu et al. [46] employed a wavelet discriminator, which employs discrete wavelet transform to obtain the HF information and then examines the frequency discrepancy to suppress the artifacts. Conde et al. [47] proposed a specific loss based on HF information, enabling the prediction to restore similar HF details as the reference. These approaches employ various techniques to extract HF information from images or deep features, illustrating that incorporation of HF details can effectively enhance the sharpness and overall quality of the SR outcomes. However, these methods for extracting HF information are rather limited and can not adequately capture the diverse inherent attributes of HF data from MR images.

3. Methodology

In this section, we first describe the problem formulation of SISR and then provide the details of two major modules. Finally, we present the loss function of our training process.

3.1. Problem statement

Let $\boldsymbol{I}_{HR} \in \mathbb{R}^{H\times W}$ and $\boldsymbol{I}_{LR} \in \mathbb{R}^{\frac{H}{\alpha}\times\frac{W}{\alpha}}$ be the HR and LR images, where $\alpha > 1$ is downsample factor and the scanning time of $\boldsymbol{I}_{LR}$ is much less than $\boldsymbol{I}_{HR}$ in T2-weighted MRI. For SISR task, a deep learning model is employed to take $\boldsymbol{I}_{LR}$ as input and restore a super-resolution image $\boldsymbol{I}_{SR}$ similar to $\boldsymbol{I}_{HR}$. However, the public available datasets do not provide $\boldsymbol{I}_{LR}$. As a result, $\boldsymbol{I}_{HR}$ is down-sampled to generate synthetic LR images $\boldsymbol{I'}_{LR}$, which are used as inputs of the SISR networks. This down-sample algorithm $\mathcal{F}^{-1}$ can be formulated as:

$$\boldsymbol{I'}_{LR} = \mathcal{F}^{-1}(\boldsymbol{I}_{HR}) = \phi(\boldsymbol{I}_{HR}) + \varsigma \approx \boldsymbol{I}_{LR}, \tag{1}$$

where ϕ is the blurring, noise and down sampling operation and ς represents the system noise. In SISR process, $\mathcal{F}$, inverse function of $\mathcal{F}^{-1}$, is fined to recover image detail from $\boldsymbol{I'}_{LR}$.

SISR is inherently an ill-posed inverse problem. Calculating an exact, closed-form solution is impractical. Instead, the approach is to seek an optimal function f that approximates the ideal inverse solution $\mathcal{F}$ through data mapping. The process is expressed as follows:

$$\boldsymbol{I}_{SR} = f(\boldsymbol{I'}_{LR},\ \theta) \approx \boldsymbol{I}_{HR}, \tag{2}$$

where θ denotes the model parameters. In learning-based paradigm, a loss function $\mathcal{L}$ is selected to evaluate the inconsistency between $\boldsymbol{I}_{SR}$ and $\boldsymbol{I}_{HR}$ to update the network. Generally, the parametric approximate inverse function $f(x;\theta)$ is learned by solving:

$$\underset{\theta\in\Theta}{argmin} \sum (\mathcal{L}(\boldsymbol{I}_{HR},\ f(\boldsymbol{I'}_{LR},\ \theta)) + \mathcal{RE}(\theta)), \tag{3}$$

where Θ is the possible parameter set and the $\mathcal{RE}(\theta)$ is a regularization function.

3.2. Down-sample $\mathcal{F}^{-1}$ and HF map extractor (HFE)

In this research, we construct the down-sample algorithm $\mathcal{F}^{-1}$ and HF information extractor (HFE) through 2D discrete Fourier Transform (2DDFT). In the MRI reconstruction process, 2DDFT is employed to separate and reconstruct the frequency-encoded and phase-encoded signal components. 2DDFT is well-suited for extracting frequency information in MR images compared to alternative methods discussed in the **Related Work** section.

We first convert $\boldsymbol{I}_{HR}$ into k-space by 2DDFT, the coordinates of output $\mathcal{K}_{HR} \in \mathbb{C}^{H\times W}$ correspond to the spatial positions in $\boldsymbol{I}_{HR}$, while the value of each point reflects the contribution of frequency components at that position. According to the properties of $\mathcal{K}_{HR}$, the region near the center point of $\mathcal{K}_{HR}$ represents the low-frequency components and corresponds to the basic semantic information of $\boldsymbol{I}_{HR}$ [48], we crop the center information of the $\mathcal{K}_{HR}$, that can be expressed as:

$$\mathcal{K}_{LR}(m,n) = \mathcal{K}_{HR}(m + \frac{H(\alpha-1)}{2\alpha}, n + \frac{W(\alpha-1)}{2\alpha}), \tag{4}$$

where $m \in \{0, 1, \cdots, \frac{H}{\alpha} - 1\}$, $n \in \{0, 1, \cdots, \frac{W}{\alpha} - 1\}$ and $\mathcal{K}_{LR} \in \mathbb{C}^{\frac{H}{\alpha}\times\frac{W}{\alpha}}$. Then, two-dimensional inverse discrete Fourier transform (2DIDFT) is utilized to reconvert the low-frequency signal $\mathcal{K}_{LR}$ to the LR image $\boldsymbol{I'}_{LR}$. This LR image is similar as $\boldsymbol{I}_{LR}$, serves as input of SISR networks. Fig. 3 illustrates the process of generating synthetic LR images. This process produces low-resolution MRI that closely approximates those naturally generated by MR scanners, adhering to the methodologies in [49,50].

In order to extract the HF information from an image, we build the high-frequency map extractor (HFE). The HF information of $\boldsymbol{I}_{HR}$ encompasses areas with significant variations in gray values, which means edges, noise, and fine details. In k-space, it corresponds to the vicinity of the boundary of the $\mathcal{K}_{HF} \in \mathbb{C}^{H\times W}$ [48]. So, the HF information is extracted from $\mathcal{K}_{HF}$ by the following formula:

$$\mathcal{K}_{HF}(u,v) = \begin{cases} 0 & \text{if } u \in U, v \in V; \\ \mathcal{K}_{HR}(u,v) & \text{else,} \end{cases} \tag{5}$$

where U and V are discrete sets of coordinates, $U = \{\frac{H(\alpha-1)}{2\alpha}, \cdots, \frac{H(\alpha+1)}{2\alpha} - 1\}$, $V = \{\frac{W(\alpha-1)}{2\alpha}, \cdots, \frac{W(\alpha+1)}{2\alpha} - 1\}$. The Eq. (5) sets the value in the central area of $\mathcal{K}_{HR}$ to zero, preserving only the portion near the four edges

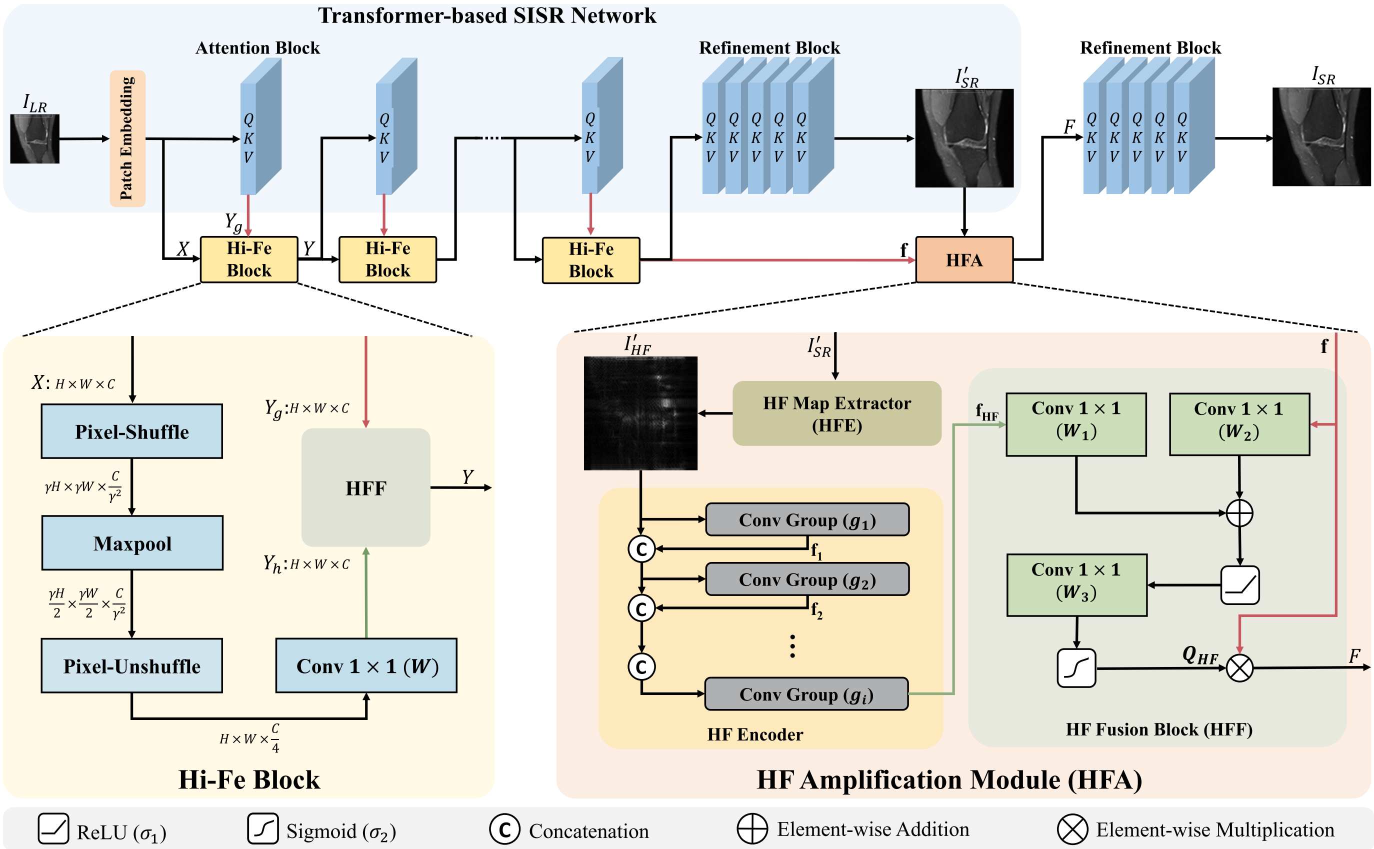

Fig. 2. The framework of our proposed module: Hi-Fe block and HFA. Each attention block in the Transformer-based SISR network is paired with a Hi-Fe Block, comprising the HF branch (blue) and the HF fusion block (HFF, green). The HFA module appended to the end of the backbone network includes three components: HF Map Extractor (HFE, tawny), HF encoder (organ), and HFF (green). Note that the HFF is present in both the Hi-Fe block and HFA modules, with its two input directions denoted by the red and green arrows.

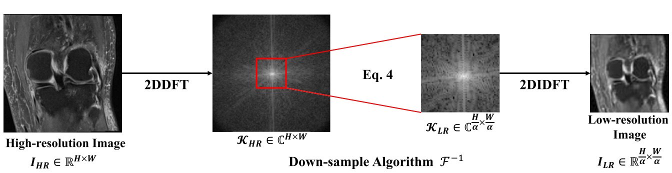

Fig. 3. Illustration of synthetic LR image generation through $\mathcal{F}^{-1}$, which consists of three steps: 2DDFT, center cropping (Eq. (4)), and 2DIDFT.

as $\mathcal{K}_{HF}$. Then, $\mathcal{K}_{HF}$ was transformed to $\boldsymbol{I}_{HF} \in \mathbb{R}^{H\times W}$ by 2DIDFT, as depicted in Fig. 1(b). Overall, HFE employ 2DDFT, Eq. (5), and 2DIDFT to capture HF information from MR images, which plays a vital role in SISR tasks.

3.3. High-frequency block (Hi-Fe block)

As discussed in Section 1, Transformer-based methods often struggle with excessive enhancement of low-frequency information and underperform in recovering HF information.

To address this challenge, we propose Hi-Fe block, a simple yet effective module to extract and integrate high-frequency information in deep feature space. The pipeline of Hi-Fe block is depicted in the lower left corner of Fig. 2, the blue pipe is HF branch, and the green rectangle is the high-frequency fusion block (HFF), which will be described in section 3.4. Instead of simply feeding image embeddings to the attention block ($Attention$), the HF branch is responsible for the extraction of HF features through max-pooling operation and parallel convolution layer.

Technically, the input feature map $X \in \mathbb{R}^{H\times W\times C}$ will be input in parallel with both the HF branch and the attention block. We first adopt $Attention$ from the backbone network as the main structure to communicate global information among X. This branch is defined as $Y_g = Attention(X)$, where $Y_g \in \mathbb{R}^{H\times W\times C}$ is the output of the attention block, representing the global presentation in deep feature space. Then, we propose the HF branch to build a reference standard identifying the portion of the input feature map X that aligns with the HF information. This branch includes maxpool operation, pixel-shuffle [51] (PS), pixel-unshuffle (PUS), and convolution layer (W). The HF Branch is expressed as:

$$Y_h = W(PUS(MaxPool(PS(X)))), \tag{6}$$

where $Y_h \in \mathbb{R}^{H\times W\times C}$ denotes the output of the HF branch, representing the HF portion in X.

Note that in shallower layers, there is a gradual gathering of visual information to achieve a global understanding of the input, while as the network deepens, it increasingly focuses on modeling high-frequency details [52]. So, we design a sample decrement structure that regulates the flow of channel features during max-pooling operations, aiming to retain a greater amount of HF information in the lower layers. The upscaling and downscaling factors (denoted as r) in pixel-shuffle and pixel-unshuffle, along with the kernel size and padding parameters in max-pooling operations, exhibit a gradual reduction as image embeddings move from the top to the bottom layers. Thus, with the different proportions of sampling and receptive fields of maxpool, the HF Branch can effectively trade off HF and LF information among all layers.

Finally, HF reference template Y_h and global feature Y_g will be fed into HFF to each transformer layer output Y, it is formulated as: $Y = HFF(Y_g, Y_h)$.

3.4. HF amplification module (HFA)

To generate faithful and detailed SR images, we propose the HFA module to extract and amplify HF details from the image intensity space, such that clear structures and rich texture details can be fully recovered. The HFA schema is presented in Fig. 2, we can clearly see that: 1) HFA comprises three distinct components: the HF map extractor (HFE), the HF encoder, and the high-frequency fusion block (HFF); 2) HFA takes two input data sources: $\boldsymbol{I}'_{SR} \in \mathbb{R}^{H\times W}$, representing the coarse output from the backbone network; $\mathbf{f}$, denoting the feature that precedes the final refinement block group of the backbone.

Unlike the Hi-Fe block, HFA operates within the intensity space. Firstly, we utilize HFE (mentioned in Section 3.2) to extract the HF map $\boldsymbol{I}'_{HF} \in \mathbb{R}^{H\times W}$ from $\boldsymbol{I}'_{SR}$, which serves as reference during the amplifying process. Subsequently, we compute the $\mathbf{f}_{HF}$ from HF map using HF encoder, as it is more efficient in identifying features with higher similarity among the encoded information [53]. As shown in Fig. 2, we conduct feature encoding through a multi-layer and multi-scale convolutional structure, HF encoder is formulated as follows:

$$\begin{cases} \mathbf{f}_1 = g_1(\boldsymbol{I}'_{HF}), \\ \mathbf{f}_2 = g_2(cat(\boldsymbol{I}'_{HF}, \mathbf{f}_1)), \\ \vdots \\ \mathbf{f}_{HF} = g_i(cat(\boldsymbol{I}'_{HF}, \mathbf{f}_1, \cdots, \mathbf{f}_{i-1})), \end{cases} \tag{7}$$

where i represents the layer ordinal, and g_i denotes the i layer convolution group including two sets of convolution layers, batch-normalization, and ReLU activation function. The feature of first layer, $\mathbf{f}_1$, is derived from $\boldsymbol{I}'_{HF}$ via g_1 and is subsequently concatenated with $\boldsymbol{I}'_{HF}$ along the *Channel* dimension before being input into the next convolution group g_2. After i iterations of calculations, the feature embeddings $\mathbf{f}_{HF}$ of HF map are obtained. This feature encoding strategy not only captures deep semantic features from the HF information, but also preserves relevant information from adjacent shallow pixels.

In the final stage, HFF enforces the network to grab corresponding features. We utilize a coordinate coefficient $Q_{HF} \in [0, 1]$ to prune feature response. We concatenate $\mathbf{f}$ and $\mathbf{f}_{HF}$ after their respective convolution block (W_1 and W_2) and then calculate Q_{HF} through ReLU ($\sigma 1$), 1*1*1 convolution (W_3) layer and Sigmoid($\sigma 2$) operation. Thus Q_{HF} can identify HF regions as salient parts of image and preserve only the activation relevant to the HF information amplifying task. The coordinate coefficient Q_{HF} can be calculated by:

$$Q_{HF} = \sigma_2(W_3^T(\sigma_1(W_1^T\ \mathbf{f}_{HF} + b_1 + W_2^T\ \mathbf{f} + b_2)) + b_3). \tag{8}$$

By performing element-wise multiplication between $\mathbf{f}$ and the coordinate coefficient Q_{HF}. The HF semantic information in $\mathbf{f}$ can be effectively amplified, yielding the HF enhanced features F, which are subsequently fed into the refinement module to render the final result $\boldsymbol{I}_{SR}$.

In addition, HFA can be connected to the tail of most SISR networks for the purpose of improving the quality of result images. It must be noted that the structure of the refinement block will vary depending on the choice of the backbone networks. Typically, the refinement block adopts the same structure as the final layer of the backbone network.

3.5. Loss function

For the training of any network incorporating the Hi-Fe block and HFA, $\boldsymbol{I}_{SR}$, $\boldsymbol{I}'_{SR}$ and $\boldsymbol{I}'_{HF}$ are all need to be supervised. $\boldsymbol{I}_{SR}$ and $\boldsymbol{I}'_{SR}$ are calculated in reference to the original high-resolution image $\boldsymbol{I}_{HR}$. The loss function used between $\boldsymbol{I}_{SR}$&$\boldsymbol{I}_{HR}$ and $\boldsymbol{I}'_{SR}$&$\boldsymbol{I}_{HR}$ is identical to that utilized by backbone network. The L1 pixel loss is employed as reconstruction loss to enhance HF details, comparing $\boldsymbol{I}'_{HF}$ with $\boldsymbol{I}_{HF}$, which is extracted from $\boldsymbol{I}_{HR}$ using HFE. So, the total loss function is:

$$\mathcal{L}_{total} = \lambda_{HR}\mathcal{L}_{\boldsymbol{I}_{SR}} + \lambda_{HR}\mathcal{L}_{\boldsymbol{I}'_{SR}} + \lambda_{HF}\mathcal{L}_{\boldsymbol{I}'_{HF}} \tag{9}$$

where the hyper-parameters maintain a relationship: $2 * \lambda_{HR} + \lambda_{HF} = 1$.

4. Experiments

4.1. Experimental settings

Datasets. Experiments are conducted on one publicly available dataset and two private clinical datasets. **fastMRI**: We randomly selected 960 knee magnetic resonance images of FS-PD volumes in the fastMRI dataset [54] provided by New York University School of Medicine. **Clinical brain MRI dataset**: This dataset was acquired using a 3T Philips Ingenia MRI system (Philips Healthcare, Best, the Netherlands) scanner. We sampled 763 T2-FLAIR full k-space acquisition data from 30 healthy subjects for the in-house brain dataset (with acquisition parameters set to TE: 120 ms). **Clinical pelvic MRI dataset**: we randomly selected a subset of 1920 pelvic T2-weighted MR images from the entire dataset. We only chose T2-modality and FS-PD images for evaluation, which is more in line with clinical practical value. All participants in both in-house datasets provided informed consent for inclusion in the study, which was conducted with approval from the local institutional review board in accordance with the principles of the Declaration of Helsinki. The MRI scanning has been approved by the institutional review board. The data is partitioned into train, valid, and test datasets as 640\160\160, 510\128\128, and 1280\320\320 separately.

Preprocessing. Following [49], we adopt the following pre-processing steps: firstly, all images are reshaped into 256×256, serving as HR images, by cropping in k-space. Then we obtain input LR images with the size of 128×128 and 64×64 through downsampling in the frequency domain [55,50]. For the down-sampling factors 2× and 4×, the central 25% and 6.25% data points are kept. Finally, we apply the inverse Fourier transform to transform the altered data into the image domain, generating the LR image. Note that all MR images data simulate 2-channel complex-valued images, and we separated real and imaginary parts and stake them as two-channel input.

Comparison Methods. We evaluate the HF information restoration capabilities of Hi-Fe block and HFA by incorporating them into eight different open-source state-of-the-art super-resolution networks, including CSN [33], ESRT [39], ENCLN [34], ELAN [56], SwinIR [26], Omni-SR [38], DAT [57], and Restormer [24]. We conduct experiments under two different scenarios: (1) Scenario 1: we assess the modules' generalizability and HF restoration performance by incorporating the two proposed modules into diverse backbone networks with upsampling factors of 2× and 4×. (2) Scenario 2: we evaluate the Hi-Fe block and HFA respectively, based on the SOTA SISR network Restormer [24] and ELAN [56] with upsampling factors of 2× and 4× as ablation study.

Implementation Details. All networks are trained using the Adam optimizer [58] with an initial learning rate of $1e-4$ and the L2 regularization function we used with the weight decay of $1e-4$. We employ a CosineAnnealing learning rate scheduler, which gradually reduces the learning rate to $4 * 10^{-8}$ during the final communication round of training. The batch size varies across different datasets: it is set to 4 for FastMRI, and 2 for both brain and pelvis MRI datasets, while we ensure that the total number of iterations remains around 100,000. We conducted all experiments on NVIDIA Tesla V100 GPUs with memory of 32 GB. The networks are implemented with PyTorch 1.7.0. Official implementations for each backbone network are employed to ensure fair comparisons. The hyperparameters of the comparison methods are kept consistent with their original settings.

Evaluation Metrics. To quantitatively evaluate the performance of the proposed modules, we utilize two commonly used metrics: peak signal-to-noise ratio (PSNR) and structural similarity index measure

Table 1
Quantitative comparison of super-resolution images between original backbones and their counterparts optimized by our proposed modules on UF = 2,4.

UF	Method	fastMRI		Brain		Pelvic	
		PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑
2×	CSN	25.27	0.7537	30.85	0.9260	34.63	0.9666
	CSN+Ours	28.65	0.8403	31.03	0.9354	34.84	0.9647
	ESRT	30.75	0.8682	32.80	0.9597	35.54	0.9728
	ESRT+Ours	30.98	0.8724	33.04	0.9607	**36.62**	0.9784
	ENCLN	29.66	0.8562	33.39	0.9527	34.92	0.9673
	ENCLN+Ours	30.71	0.8670	33.56	0.9596	35.12	0.9745
	ELAN	31.77	0.9455	31.83	0.9314	35.08	0.9671
	ELAN+Ours	**32.17**	**0.9496**	32.17	0.9425	35.27	0.9690
	SwinIR	27.19	0.8135	33.27	0.9601	35.24	0.9711
	SwinIR+Ours	28.43	0.8478	**34.41**	**0.9765**	35.76	**0.9803**
	Omni-SR	30.52	0.8579	32.13	0.9550	34.16	0.9643
	Omni-SR+Ours	30.79	0.8621	32.86	0.9571	35.01	0.9681
	DAT	30.89	0.8698	32.60	0.9612	35.40	0.9721
	DAT+Ours	31.23	0.8719	33.30	0.9638	35.78	0.9733
	Restormer	30.59	0.8646	32.02	0.9511	34.90	0.9689
	Restormer+Ours	31.68	0.8753	33.24	0.9645	35.64	0.9791
4×	CSN	24.15	0.6727	26.82	0.8130	29.17	0.8891
	CSN+Ours	25.38	0.6968	27.57	0.8545	29.86	0.9062
	ESRT	25.43	0.7335	27.47	0.8365	29.43	0.9078
	ESRT+Ours	27.11	0.7406	27.82	0.8680	29.96	0.9101
	ENCLN	27.43	0.7548	27.55	0.8814	29.85	0.9186
	ENCLN+Ours	27.50	0.7545	27.78	0.8912	30.32	0.9214
	ELAN	24.41	0.7057	27.29	0.8434	29.53	0.9049
	ELAN+Ours	25.86	0.7100	27.53	0.8657	29.96	0.9106
	SwinIR	27.76	0.7537	27.26	0.8782	29.89	0.9204
	SwinIR+Ours	27.92	0.7615	27.74	**0.8971**	30.27	0.9212
	Omni-SR	26.44	0.7259	27.21	0.8593	29.36	0.9048
	Omni-SR+Ours	27.84	0.7369	27.40	0.8657	29.64	0.9216
	DAT	25.98	0.7302	26.38	0.8884	29.96	0.9202
	DAT+Ours	26.33	0.7434	26.65	0.8806	30.17	**0.9223**
	Restormer	27.90	0.7542	27.10	0.8778	29.39	0.9131
	Restormer+Ours	**28.38**	**0.7623**	**28.36**	0.8820	**30.40**	0.9208

(SSIM). These two metrics are used to evaluate the image quality on two pairs of images: $\boldsymbol{I}_{HR}\&\boldsymbol{I}_{SR}$ and $\boldsymbol{I}_{HF}\&\boldsymbol{I'}_{HF}$. The larger PSNR and SSIM values represent the better super-resolution performance. Apart from quantitative comparisons, we also present visualizations of error maps from both $\boldsymbol{I}_{HR}$ and $\boldsymbol{I}_{SR}$, which serve to illustrate the performance disparities among different methods. In the error map, regions with brighter intensity levels correspond to areas with higher error.

4.2. Results

To evaluate the effectiveness of our proposed Hi-Fe block and HFA module, we integrate these two modules into 6 different networks. We compare the performance on MR SISR and the corresponding HF map on two different upsample factors (UF = 2, 4) against the original networks. The experimental results on three different MRI datasets are presented in Table 1, and Table 2.

For MR SISR results, presented in Table 1, it is evident that all methods exhibit better super-resolution performance at UF = 2, but their performance diminishes significantly at larger upsample factors. After integrating our proposed Hi-Fe block and HFA module, all methods exhibit improvements in performance at UF = 4. Among them, CSN and ESRT showcase the most significant PSNR enhancement on the fastMRI dataset, from 24.15 to 25.38 and 25.43 to 27.11 respectively. For individual backbone networks, we can see that the Restormer does not achieve the best results on the Pelvic and Brain datasets. However, the incorporation of our modules notably boosts Restormer's performance, achieving a PSNR value of 23.24 and an SSIM of 0.4426, thereby outshining the outcomes of other networks, even those similarly enhanced with our modules. This effectively demonstrates that Hi-Fe block and HFA can enhance the performance of Transformer-based networks, aligning with the viewpoint discussed before.

The data analysis from Table 2 corroborates the same conclusion with respect to the HF map results. Especially for the Brain dataset, the enhancement through our modules has led to a significant improvement in Restormer's performance in HF map restoration. For instance, at a 4× magnification, Restormer+Ours achieves the highest PSNR value of 25.07 and an SSIM of 0.6690, markedly surpassing the SR results of other backbone networks. These results demonstrate that our proposed modules significantly enhance the super-resolution performance of various networks, particularly in terms of restoring HF information.

To better analyze the improvement of our modules for MR SISR, we visualize some cases with UF = 4, as shown in Fig. 4. The visual results of **Ours** are produced by integrating our proposed modules with the SOTA SISR network, Restormer. The results from all experimental methods do not present obvious anatomical structure differences, owing to the exceptional repair capabilities of these backbone networks for LF information, which is crucial in defining the basic structure of the image. While our proposed modules are specifically designed to restore HF information of anatomical structure boundaries, and the resulting visualizations align closely with the intended purpose of these modules as previously described. Due to the relatively small differences between the MR super-resolution images, we also provide corresponding error maps and HF maps to further elucidate the effectiveness of our modules. The textures in the HF map represent the HF information corresponding

Table 2
Quantitative comparison of HF maps between original backbones and their counterparts optimized by our proposed modules on UF = 2,4.

UF	Method	fastMRI		Brain		Pelvic	
		PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑
2×	CSN	22.03	0.3503	31.73	0.8974	34.64	0.9668
	CSN+Ours	23.72	0.5715	31.88	0.9068	34.84	0.9648
	ESRT	24.82	0.6232	32.41	0.9526	34.97	0.9676
	ESRT+Ours	25.21	0.6292	32.82	0.9598	35.26	0.9734
	ENCLN	25.29	0.6406	33.10	0.9533	35.41	0.9723
	ENCLN+Ours	25.69	**0.6494**	33.42	0.9597	**35.69**	0.9748
	ELAN	24.50	0.5838	31.85	0.9487	35.18	0.9642
	ELAN+Ours	24.63	0.5964	32.21	0.9505	35.33	0.9699
	SwinIR	23.59	0.5723	33.29	0.9601	35.29	0.9714
	SwinIR+Ours	24.32	0.5812	**33.83**	**0.9682**	35.64	0.9738
	Omni-SR	24.76	0.6028	32.15	0.9552	34.19	0.9645
	Omni-SR+Ours	25.35	0.6267	32.87	0.9574	35.05	0.9683
	DAT	25.45	0.6403	32.10	0.9614	34.82	0.9627
	DAT+Ours	24.37	0.6015	32.62	0.9575	35.54	0.9732
	Restormer	25.17	0.6284	32.03	0.9513	34.95	0.9692
	Restormer+Ours	**25.71**	0.6357	32.89	0.9618	35.37	**0.9772**
4×	CSN	21.23	0.2375	21.81	0.4887	22.18	0.5968
	CSN+Ours	22.07	0.3732	22.36	0.5064	22.87	0.6877
	ESRT	22.24	0.3891	21.86	0.4823	22.10	0.5785
	ESRT+Ours	22.50	0.3905	21.88	0.4631	22.93	0.6322
	ENCLN	22.15	0.4068	22.91	0.5109	22.33	0.6201
	ENCLN+Ours	22.18	0.4090	23.29	0.5317	23.15	0.6670
	ELAN	21.35	0.3180	21.30	0.4072	21.95	0.5376
	ELAN+Ours	22.30	0.3662	22.42	0.4297	22.29	0.5798
	SwinIR	22.40	0.4254	21.12	0.3291	22.32	0.6804
	SwinIR+Ours	22.69	0.4406	22.61	0.5622	23.63	**0.7014**
	Omni-SR	22.11	0.4396	21.20	0.4352	22.36	0.6751
	Omni-SR+Ours	22.48	0.4423	22.40	0.4538	23.12	0.6823
	DAT	21.72	0.3508	21.98	0.4492	20.72	0.5300
	DAT-SR+Ours	22.30	0.3951	22.65	0.5387	21.48	0.5574
	Restormer	22.29	0.4141	22.88	0.5834	22.57	0.6203
	Restormer+Ours	**23.24**	**0.4426**	**25.07**	**0.6690**	**23.94**	0.6975

to the super-resolution images, providing an additional visual demonstration of our module's restoration capabilities in the HF domain. The red markings in the HF map indicate regions that are more similar to the ground truth compared to other methods, demonstrating that our modules significantly enhance image quality in the HF domain. The predominant textures in the error map indicate worse reconstruction quality, offering a macroscopic comparison of the network's restorative capabilities in the image intensity domain. After optimization with our modules, the error maps, when observed from a macroscopic viewpoint, predominantly show lower saturation and cooler tones across more extensive areas compared to those of other methods, signifying reduced errors. Overall, these HF maps and error maps prove the exceptional performance of our proposed module in maintaining image details and overall consistency.

4.3. Ablation study

4.3.1. Effectiveness of proposed modules

In this paper, we introduce two modules tailored to different information domains: Hi-Fe block corresponds to the deep feature space and HFA pertains to the image intensity space. To evaluate the efficacy of each module, we use two SOTA networks, Restormer and ELAN, as the baseline models and conduct ablation experiments on three MRI datasets at UF = 2 and UF = 4. The ablation experiments results are shown in Table 3. As shown in Table 3, by separately incorporating Hi-Fe block and HFA, the model's super-resolution performance is improved, with each component contributing to the enhancement individually. When Hi-Fe block is used to provide HF information in deep features of HF branch to plain Restormer attention blocks, the performance was significantly improved with average increment of 1.03% and 1.79% in PSNR, as well as 0.38% and 2.66% in SSIM, on super-resolution images and HF maps, respectively. Furthermore, solely incorporating the HFA module results in the performance improvement of 1.75% and 2.27% in PSNR, and 0.92% and 4.27% in SSIM, surpassing the performance achieved by solely using the Hi-Fe block.

This improvement is attributed to the HFA's ability to more precisely provide pixel-level HF information, thus more effectively boosting network performance. The simultaneous integration of both the Hi-Fe block and the HFA module markedly elevates super-resolution performance, demonstrating a synergistic effect that results in substantial gains of 3.21. The experimental results on two SOTA Transformer-based network reveal that the combined use of Hi-Fe block and HFA significantly outperforms the individual application of either component in terms of PSNR and SSIM.

The goal of the Hi-Fe block is to moderately enhance regions within the deep feature space that exhibit significant anatomical structural texture variations. While it can restore most areas with pronounced texture differences effectively, there exists a mapping relationship from high dimension to low dimension, which makes it challenging to repair regions with substantial texture variations. By individually adding the HFA module, we can extract texture features from the HF map at the same resolution as the MR images to facilitate a better recovery of anatomical structural edges. Although it can enhance the significance of anatomical structures with a large pixel scale in the model, for smaller tissues, the minimal differences they exhibit in comparison to the surrounding areas can result in the loss of texture information during HFE

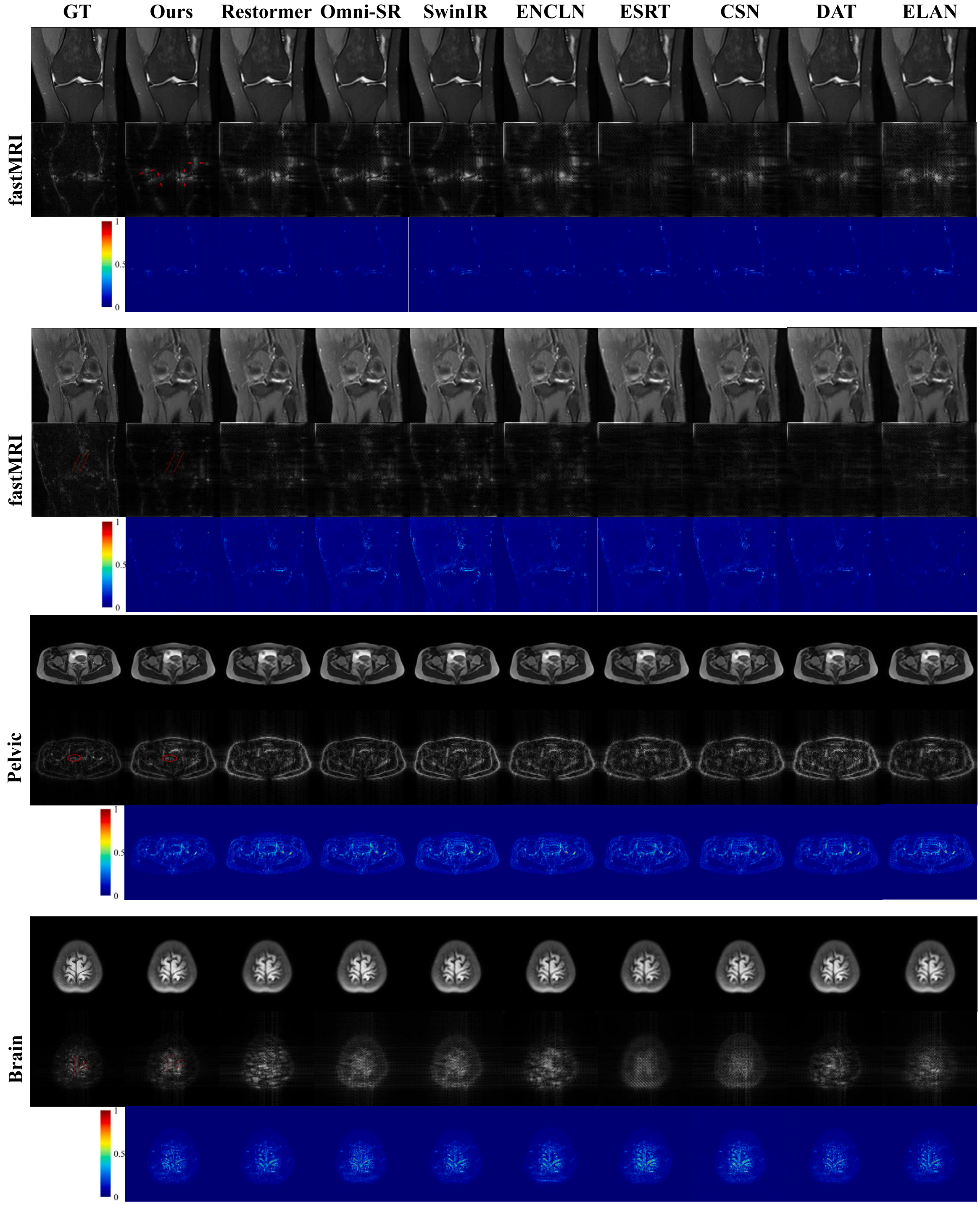

Fig. 4. Visual comparison of the super-resolution results, their corresponding HF map, and error map among different methods. The results of **Ours** are generated by combining our modules (Hi-Fe blocks and HFA module) into Restormer. The red color highlights the regions that are more similar to ground truth. Evidently, the visual outcomes of **Ours** reveal more specific HF structures in the HF maps, and concurrently demonstrate reduced discrepancies in the error map, signifying superior restoration of high-quality HF information and lower extent of difference.

sampling. Therefore, the combination of the Hi-Fe block and the HFA can address the aforementioned issues and further enhance the model's performance by clustering HF features in high-dimensional space and boosting HF texture regions.

4.3.2. Comparison of different HF extraction methods

To validate the superiority of 2DDFT in extracting HF information for MR image SISR tasks, we conduct experiments to investigate the effects of various HF information extraction methods on the quality of the resulting super-resolution images. This experiment was conducted using ELAN+Ours as the backbone, with an UF of 4 on the fast MRI dataset. We compared two most representative methods of HF information extraction: two-dimensional discrete wavelet transform (2DDWT) and Gaussian filter (GF).

For the 2DDWT, we employed the PyTorch-based DTCWT function described in [59]. We set the hyperparameter $\mathcal{J}$ in the DTCWT to 1, which generates three HF detail maps of size 128x128 in horizontal, vertical, and diagonal directions. Subsequently, after summing three HF

Table 3
Quantitative comparison of result images and HF maps from ablation study based on Restormer on three datasets.

UF	Method	fastMRI		Brain		Pelvic	
		PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑
2×	Restormer	30.59	0.8646	32.02	0.9511	34.90	0.9689
	Restormer + Hi-Fe block	30.96	0.8675	32.57	0.9564	35.12	0.9724
	Restormer + HFA	31.17	0.8684	32.76	0.9602	35.57	0.9768
	Restormer + Ours	**31.68**	**0.8753**	**33.24**	**0.9645**	**35.64**	**0.9791**
	ELAN	31.83	0.9421	31.83	0.9314	35.08	0.9671
	ELAN + Hi-Fe block	31.95	0.9461	31.99	0.9358	35.16	0.9674
	ELAN + HFA	32.03	0.9470	32.06	0.9411	35.24	0.9686
	ELAN + Ours	**32.17**	**0.9496**	**32.17**	**0.9425**	**35.27**	**0.9690**
HF map 2×	Restormer	25.17	0.6284	32.03	0.9513	34.95	0.9692
	Restormer + Hi-Fe block	25.34	0.6311	32.21	0.9547	35.02	0.9748
	Restormer + HFA	25.63	0.6334	32.72	0.9582	35.24	0.9759
	Restormer + Ours	**25.71**	**0.6357**	**32.89**	**0.9618**	**35.37**	**0.9772**
	ELAN	24.50	0.5838	31.85	0.9487	35.18	0.9642
	ELAN + Hi-Fe block	24.54	0.5872	31.99	0.9496	35.24	0.9641
	ELAN + HFA	24.59	0.5947	32.15	0.9500	35.27	0.9645
	ELAN + Ours	**24.63**	**0.5964**	**32.21**	**0.9505**	**35.33**	**0.9699**
4×	Restormer	27.90	0.7542	27.10	0.8778	29.39	0.9131
	Restormer + Hi-Fe block	28.27	0.7606	27.91	0.8759	29.99	0.9178
	Restormer + HFA	28.25	0.7586	28.02	**0.8942**	30.24	0.9205
	Restormer + Ours	**28.38**	**0.7623**	**28.36**	0.8820	**30.40**	**0.9208**
	ELAN	24.41	0.7057	27.29	0.8434	29.53	0.9049
	ELAN + Hi-Fe block	24.57	0.7065	27.33	0.8517	29.70	0.9065
	ELAN + HFA	25.35	0.7074	27.48	0.8612	29.83	0.9079
	ELAN + Ours	**25.86**	**0.7100**	**27.53**	**0.8657**	**29.96**	**0.9106**
HF map 4×	Restormer	22.29	0.4141	22.88	0.5834	22.57	0.6203
	Restormer + Hi-Fe block	22.98	0.4325	24.53	0.6100	23.61	0.6827
	Restormer + HFA	23.17	0.4417	23.99	0.6468	23.45	0.6671
	Restormer + Ours	**23.24**	**0.4426**	**25.07**	**0.6690**	**23.94**	**0.6975**
	ELAN	21.35	0.3180	21.30	0.4072	21.95	0.5376
	ELAN + Hi-Fe block	21.84	0.3364	21.64	0.4168	22.06	0.5527
	ELAN + HFA	22.16	0.3596	22.07	0.4231	22.14	0.5648
	ELAN + Ours	**22.30**	**0.3662**	**22.42**	**0.4297**	**22.29**	**0.5798**

Table 4
Quantitative comparison of three HF information extraction methods on ELAN+Ours, UF = 4 and fast MRI dataset.

Method	PSNR↑	SSIM↑
2DDFT	**25.86**	**0.7100**
2DDWT	25.26	0.7079
Gaussian Filter	25.62	0.7047

maps, we upsampled the combined maps by a factor of two using linear interpolation, and then replaced the HF maps required in the HFA with these upsampled maps. For the GF, we first extract LF information from the super-resolution image using a 2D convolution with a kernel size of 3x3. The convolution kernel is as follows:

$$\begin{bmatrix} \frac{1}{16} & \frac{1}{8} & \frac{1}{16} \\ \frac{1}{8} & \frac{1}{4} & \frac{1}{8} \\ \frac{1}{16} & \frac{1}{8} & \frac{1}{16} \end{bmatrix}. \tag{10}$$

By employing the GF to produce a LF image and then deducting this from the super-resolution image, we obtain the final HF maps. The above two methods of HF extraction follow the practice in [45,46].

The experimental results are shown in Table 4. It is obvious that the HF information extraction method based on 2DDFT yields the best super-resolution images compared to the other two methods. This affirms our default setting: 2DDFT excels in extracting frequency information from MR images, demonstrating superior performance over alternative approaches.

4.4. LF maps analysis

We also investigate the potential effects of our proposed modules on LR recovery. Specifically, we extracted LF maps based on experimental results from five representative backbones: Restormer, SwinIR, Omni-SR, ESRT, and ENCLN in fastMRI UF = 2. This comparison assesses whether integrating our module also enhances the faithful reconstruction of LF information. We referred to the Down-Sample $\mathcal{F}^{-1}$ described in Section 3.2 to extract LF maps. For evaluating the high-resolution reconstruction of LF information, we preserve $\mathcal{K}_{LR}$ at the center of $\mathcal{K}_{HR}$, set the remaining areas as 0, and subsequently execute 2DIDFT. This procedure generates LF maps with dimensions $H \times W$, encapsulating LF information in high-resolution image.

The results are presented in Table 5. We can observe that across three distinct datasets, the average increases in PSNR and SSIM were 3.32% and 0.61%, respectively. These findings suggest that, despite the primary optimization of our module for high-frequency information, the comprehensive supervision strategy employed by our network also effectively enhanced the reconstruction of low-frequency information in super-resolution images.

4.5. Complexity analysis

The comparison of parameters numbers, flops and runtime of different methods in fastMRI UF = 4 SISR is described in Table 6. Table 6 presents both parameter number, and runtime among various backbones and the counterpart with integration of our modules. We can see that our modules incur an acceptable increase of parameter number and

Table 5
Quantitative comparison of LF maps between original backbones and their counterparts optimized by our proposed modules on UF = 2.

UF	Method	fastMRI		Brain		Pelvic	
		PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑
2×	Restormer	39.96	0.9868	41.17	0.9884	46.48	0.9960
	Restormer+Ours	40.04	0.9885	44.46	0.9920	47.45	0.9963
	SwinIR	29.74	0.9284	46.50	0.9949	49.63	0.9975
	SwinIR+Ours	33.48	0.9702	47.62	0.9963	50.21	0.9980
	Omni-SR	40.39	0.9853	41.00	0.9865	45.78	0.9944
	Omni-SR+Ours	39.95	0.9860	39.85	0.9875	47.04	0.9963
	ESRT	40.21	0.9911	44.24	0.9906	46.20	0.9957
	ESRT+Ours	41.03	0.9989	42.77	0.9893	47.04	0.9957
	ENCLN	40.55	0.9893	40.27	0.9864	49.88	0.9976
	ENCLN+Ours	40.17	0.9859	40.37	0.9898	49.38	0.9976

Table 6
Complexity Analysis of different methods.

Method	Param.(M)	Time (ms)	Method	Param.(M)	Time (ms)
CSN	6.96	32.33	SwinIR	11.66	257.61
CSN+Ours	12.61	89.94	SwinIR+Ours	12.01	306.09
ESRT	0.75	157.00	Omni-SR	2.79	100.48
ESRT+Ours	0.92	211.70	Omni-SR+Ours	3.00	148.17
ENLCN	1.83	21.41	DAT	4.86	107.83
ENLCN+Ours	3.78	37.18	DAT+Ours	5.12	135.09
ELAN	0.57	102.16	Restormer	5.65	122.67
ELAN+Ours	0.89	115.64	Restormer+Ours	13.64	164.22

running time. It proved that our plug-and-play modules are efficient to achieve better super-resolution performance.

4.6. User study

This study involved 10 exceptionally skilled and board-certified radiologists, with a minimum of 8 years of extensive experience. Each radiologist was assigned 36 pairs of images for detailed examination, with each pair consisting of one image produced by a backbone SISR network and its counterpart enhanced by our advanced module. For comparative purposes, each of these backbone networks contributed six pairs of images. Then the radiologists judiciously selected the visually superior image from each pair. The final percentage epitomizes the collective average preference from all users across all images. As shown in Fig. 5, more than 87.5% of the users distinctly favored the images precisely crafted after careful optimization with our Hi-Fe block and HFA module.

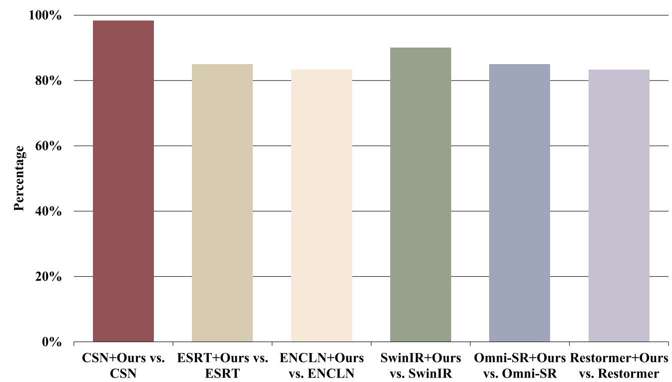

Fig. 5. The user study results of different SISR methods and our modules.

5. Discussion

We evaluated our modules on three MR image datasets. Single image super-resolution remains highly challenging, primarily owing to complicated anatomical structures and the heterogeneity as well as diversity found in MR images. It is extremely difficult for imperceptible and complicated structures, especially adhesion areas between tissue structures, which can be represented as the HF components of images. Compared to other SOTA approaches, general SISR networks, with the enhancement of our HFA and Hi-Fe block, not only achieved a statistically overall improvement with respect to the metrics of PSNR and SSIM, but also exhibited visually noticeable overall enhancements in repairing adhesion areas between tissue structures. Note that our method is two general plug-and-play modules. In addition to the backbone models discussed in this study, it can also be applied to other SISR networks.

Although with satisfactory overall restoration performance, the enhancement provided by our proposed modules under certain foundational networks still falls short of expectations. As shown in Table 1 and 2, the enhancement effect of our modules on ENCLN still requires improvement. In the cases of brain and pelvic targets with a upscale factor of 2, as presented in Tables 1 and 2, ENCLN's focus on efficiently delivering non-local information resulted in our module's inability to extract relevant HF information from the features extracted from the backbone, thereby hindering substantial overall improvements. To address these issues, we still have to optimize our modules through a comprehensive analysis of the information and feature type extracted by both the backbone network and our modules in the future.

6. Conclusion

In this paper, we propose two plug-and-play modules for MRI single image super-resolution with Transformer-based networks via high-frequency information restoration. For Hi-Fe block, we introduce key designs to the core components of the transformer attention block for perking up concentrations on high-frequency and expanding the perception capability in deep feature space. For HFA, we build a succinct structure that not only can be utilized in most SISR networks, but also amplifies high-frequency information on global context in image intensity space. To incorporate the strength of Hi-Fe block and HFA, concatenating them can cooperatively enrich the high-frequency content from feature domain and image domain. Extensive experimental findings on three datasets demonstrate that our Hi-Fe block and HFA can improve the super-resolution quality of MR images from numerous SOTA SISR networks, especially in high-frequency details.

CRediT authorship contribution statement

Haoneng Lin: Writing – original draft, Visualization, Validation, Software, Project administration, Conceptualization, Data curation, Formal analysis, Methodology. **Jing Zou:** Conceptualization, Writing – review & editing. **Kang Wang:** Conceptualization, Methodology. **Yidan Feng:** Conceptualization, Data curation. **Cheng Xu:** Writing – review & editing, Visualization, Formal analysis. **Jun Lyu:** Conceptualization, Data curation, Formal analysis, Writing – review & editing. **Jing Qin:** Formal analysis, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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